Indian Institute of Technology Ropar
School of Artificial Intelligence and Data Engineering
Deep Learning (AI-504)

AI-504
Mar 02, 2026

Time: 2 hour — *Midsem : Academic Year 2025-26* — Maximum Marks: 25

General Instructions:
- This paper has 4 questions. All are compulsory; Question 4 has an internal choice-attempt any one.
- Make suitable assumptions wherever necessary. Use of calculator is allowed.
- Usage of cell in the exam hall is strictly prohibited.
- Write the answer neatly in answer sheets. Follow same question order.

1. Answer the following questions:

(a) Consider the two activation functions, Sigmoid(x) and $\tanh(x)$. Compare them based on the agnitude of their gradients, which reflects the steepness of learning. Compute and evaluate eir derivatives at $x = 0$ and $x = 0.25$, and determine which activation function has the larger gradient at each point.

(b) Consider the vector $x = [2, -1, 2]$. [2]
   (a) Compute the 1-norm $\|x\|_1$. [2]
   (b) Compute the 2-norm (Euclidean norm) $\|x\|_2$.

(c) Write the gradient descent weight update rule. Explain the effect of very small and very large learning rates.

(d) Define TP, FP, FN, TN using confusion matrix. When is accuracy misleading? [2]

(e) Consider three classification models with the following performance: Model 1 has a training error of 2% and a test error of 25%. Model 2 has a training error of 25% and a test error of 26%. Model 3 has both training error and test error less than 10%. Based on these results, determine which model represents overfitting, underfitting, and good generalization. Briefly justify your answer by explaining the relationship between training and test errors in each case. [2]

(f) Consider the cost function: $J(\theta) = \theta^2 - 4\theta + 5$. Starting from the initial parameter value $\theta_0 = 2$ and using a learning rate $\eta = 0.1$, perform one step of gradient descent and compute θ_1. [2]

[2]

medical imaging task, you are training a neural network to classify whether a lesion is **malignant** , cancerous) or **benign** (non-cancerous) based on input features extracted from the image. The work has a single hidden layer with two neurons and one output neuron. Activation: sigmoid $\sigma(x) = \dfrac{1}{1+e^{-x}}$. Loss: Mean Squared Error (MSE) $= \frac{1}{2}(y_{\text{target}} - y_{\text{pred}})^2$. [6]

Given the following details for a single training example:

i) Input features: $x_1 = 1$, $x_2 = 1$.

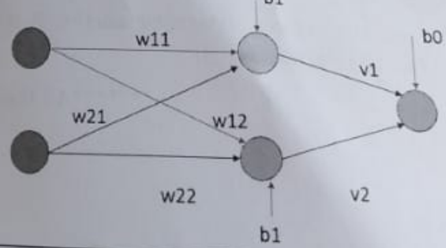
ii) Hidden-layer weights (input $\rightarrow$ hidden): $w_{11} = 0.1$, $w_{12} = -0.2$, $w_{21} = 0.3$, $w_{22} = 0.2$.

iii) Output-layer weights (hidden $\rightarrow$ output): $v_1 = 0.2$, $v_2 = -0.1$.

iv) Biases: $b_1 = 0.1$ (hidden), $b_0 = 0.05$ (output).

v) Target output: $y_{\text{target}} = 1$.

vi) Learning rate: $\eta = 0.1$.



a) Perform the forward pass and compute y_{pred} (show pre-activations and activations). [2]

b) Compute the error using the MSE. [1]

c) Using backpropagation, compute the gradients of all weights $(v_1, v_2, w_{11}, w_{12}, w_{21}, w_{22})$ and biases (b_0, b_1). [2]

d) Update the weights using gradient descent and report the updated values. [1]

3. In an autonomous driving system, you are training a model to classify frames into one of three driving events: **Pedestrian**, **StopSign**, or **Obstacle**. The model **M1** is tested on 200 frames and yields the following confusion matrix (Rows = **Predicted**, Columns = **Actual**): [4]

Predicted\Actual	Pedestrian	StopSign	Obstacle
Pedestrian	40	10	10
StopSign	5	45	10
Obstacle	5	5	70

(a) Calculate the **overall accuracy** of the model. [1]

(b) For the **Pedestrian** class, determine the **precision** and the **recall**. [1]

(c) Using the precision and recall from part (b), compute the **F1-score** for the Pedestrian class. [1]

(d) A second model **M2**, evaluated on the same data, reports for the **Pedestrian** class: Precision = 75%, Recall = 68%. Based on the F1-score for the Pedestrian class, decide which model (M1 or M2) performs better at detecting pedestrians and briefly justify your answer. [1]

4. In a manufacturing plant, you're tasked with predicting whether a machine will experience a fault during operation using logistic regression. The data includes two key features: the machine's temperature (in °C) and the vibration level (in m/s^2). [3]

The logistic regression model is defined as follows:

$$P(y=1|x) = \frac{1}{1+e^{-(w_0+w_1\cdot x_1+w_2\cdot x_2)}}$$

where: $w_0 = -0.5$ (bias); $w_1 = 0.8$ (coefficient for temp); $w_2 = 1.2$ (coefficient for vibration level)

Given the following input data for a machine:

Temperature $x_1 = 75\,°\mathrm{C}$ and **Vibration level** $x_2 = 0.4\,\mathrm{m/s^2}$

i) Calculate the probability that the machine will experience a fault. [1.5]

ii) Based on the calculated probability, predict whether the machine is likely to experience a fault (threshold = 0.5). [1.5]

OR

u are designing a perceptron model to classify cricket shots into two categories: **Boundary (class**) and **Defensive (class 0)**. Each shot is represented by two features: x = bat swing speed, y = ball contact angle. The dataset is given below:

Boundary shots (class 1):	$(2,4), (3,5), (4,6)$
Defensive shots (class 0):	$(1,2), (2,1), (3,2)$

The perceptron computes: $z = w_x \cdot x + w_y \cdot y + b$, $\quad \hat{y} = \begin{cases} 1 & \text{if } z > 0, \\ 0 & \text{otherwise.} \end{cases}$ with the current weights:

$w_x = 1, \quad w_y = -1, \quad b = 0.$

i) Compute the perceptron output for each data point and hence the classification accuracy (in %). [2]

ii) From the dataset, determine whether it is possible for a perceptron to achieve 100% accuracy. Justify your answer. [1]

*******All the Best**********